

AIR POLLUTION

In India, a debate over the benefits of gas stoves

Surprise finding of few health payoffs complicates push to replace biomass fuel

By **Vaishnavi Chandrashekhar**

On a cloudy morning in October 2024, researchers visited a small home in Karyalur, a village in south India, to take the measure of a 5-year-old girl named Dhanashree. They recorded her blood pressure, weight, and height, and tested her skills with games that involved kicking a ball or building with blocks. The reason for their visit sat in a corner: a cooking stove connected to a large red cylinder of liquefied petroleum gas (LPG).

The stove and cylinder had been given to Dhanashree's family as part of a large four-nation study, funded by the U.S. National Institutes of Health and the Bill & Melinda Gates Foundation, called the Household Air Pollution Intervention Network (HAPIN) trial. Launched in 2018, it aimed to discover whether the health of children improved in homes that replaced traditional, smoky stoves that burn biomass such as wood and dung with stoves fueled by cleaner burning LPG. Now, the results of the HAPIN trial are fueling a debate among researchers over whether India and other nations should aggressively expand the use of gas cooking stoves to curb air pollution.

Supporters of expanding LPG use say it is critical to improving air quality—especially indoor air quality—in a nation with some of the world's worst air pollution. They note that researchers have found biomass stoves used for cooking and heating produce 25% to 50% of the smog that chokes some regions of India for parts of each year. "Clean residential energy is key to delivering healthful air on multiple scales," says Chandra Venkataraman, a senior climate researcher at the Indian Institute of Technology Bombay.

Skeptics, however, point to the HAPIN trial's puzzling outcome: It found no statistically significant health benefits for children living in houses with LPG stoves. They add

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that LPG is a relatively expensive fossil fuel that is subject to global market turmoil. “If LPG can’t deliver on health outcomes, we should try something else,” says epidemiologist Thomas Clasen of Emory University, a leader of the HAPIN trial. Expanding access to electricity, ideally from renewable sources, would benefit both health and the environment, he and others say.

Biomass stoves, used by an estimated one-third of the world’s population, pose a clear threat to human health. The stoves produce fine soot particles, less than 2.5 micrometers in diameter (PM_{2.5}), that contribute to respiratory and other health problems and millions of premature deaths.

For decades, researchers have tried to design “smokeless” biomass stoves, with little success. More recently, they turned to replacing biomass fuels with natural gas (methane) or LPG (often propane or butane), especially after the World Health Organization endorsed them as clean fuels in 2014 and the United Nations made access to energy a sustainable development goal in 2015. In recent years, the World Bank has helped promote gas stoves in Africa, and countries such as Indonesia have expanded access to LPG.

Advocates of gas stoves say they are good for the climate as well as air quality. Studies by Venkataraman, for example, have found that the soot and other pollutants given off by biomass stoves are potent warming compounds. A study by the Council on Energy, Environment and Water (CEEW), a nonprofit research institute in India, also found that one LPG cylinder can replace 178 kilograms of firewood, leading to a net reduction in warming emissions due to unsustainable wood harvesting practices.

Researchers thought the HAPIN trial would make an unequivocal case for LPG’s health benefits. It enrolled some 3200 poor households in India, Guatemala, Peru, and Rwanda, including 800 pregnant women and their 800 infants. Half of the households received LPG stoves, which the researchers hypothesized would reduce pollution exposures and childhood pneumonia, and improve maternal blood pressure, infant birth weight, and growth rates. But the primary results, reported in three papers published in *The New England Journal of Medicine* in 2022 and 2024, were decidedly mixed. Median PM_{2.5} exposure dropped by two-thirds in homes with the stoves. But for birth weight, stunting, and pneumonia, there was no statistically significant difference after 18 months between children living in households using LPG and those burning biomass.

Several factors might explain the results, the researchers say. The intervention may have started too late in the gestation



India has been promoting stoves fueled by liquefied petroleum gas, such as this one in a home in Uttar Pradesh.

period—most mothers enrolled in their second trimester, and the few who enrolled earlier seemed to see better results. Benefits might also appear later; ongoing follow-up studies of children like Dhanashree could reveal long-term effects. But the most likely reason, Clasen says, is that when a population faces multiple environmental risks, as most poor communities do, a single intervention such as a clean stove may be “necessary but not sufficient” to improve health.

This point is also emphasized by health researcher Kalpana Balakrishnan, who led the HAPIN trial in India. “We could give them free LPG, but we couldn’t give them free food,” says Balakrishnan, who is dean of research at the Sri Ramachandra Institute of Higher Education and Research. “Perhaps it was naïve to think we can undo with a short intervention what they have already been exposed to, the course they were already on.”

The findings have left researchers divided over the wisdom of expanding the use of LPG. Balakrishnan says the demonstrated pollution reductions justify pushing ahead. But Clasen thinks policymakers should be wary of spending more public funds on the fuel without definitive evidence of health benefits. “The oil and gas industry like people to think that [LPG] is a virtuous fuel,” he says. But alternatives, such as expanding access to electricity, could offer broader benefits, such as providing lighting and stimulating economic activity.

Yet a large-scale “leapfrog” to electricity, especially from renewable sources, may be unrealistic in India. Although its grid is expanding—and solar is also being

promoted—the system isn’t ready to handle the increased demand, says analyst Abhishek Kar of CEEW. Instead, he sees the path followed by Ecuador as a potential model for transitioning India away from biomass. There, the government promoted subsidized LPG and then, when gas prices started to rise, scaled up the use of electric stoves by offering financial incentives.

In India, LPG use has a long way to go. In 2016, the government launched an initiative to connect 100 million poor households to subsidized fuel supplies. By 2020, 62% of households were using LPG as their primary cooking fuel, but the share is much lower in rural areas. And surveys have found that only 30% of homes cook exclusively with gas; the rest either use biomass or charcoal exclusively or alongside LPG.

This “fuel stacking” was evident in Dhanashree’s home. Anita, her mother, received free LPG during the HAPIN trial, spanning her pregnancy and Dhanashree’s first year. But after that she reverted to using mostly wood. Cooking with gas is quicker and not smoky, Anita said, but the family can’t afford to frequently refill cylinders, even at subsidized prices.

Advocates of expanding LPG use hope India’s new national air pollution plan, which could be finalized this year, will address household air pollution and help increase subsidies for gas. And Venkataraman says a government plan to boost the production of biogas from waste could also contribute to a transition away from biomass fuels. “We need a basket of fuels for rural energy transitions,” Venkataraman says. “But you have to start somewhere.” ■